Answer **all** the questions in the spaces provided.

1 Two spheres A and B of masses 3.7 kg and 2.7 kg respectively are attached to two light inextensible strings L_1 and L_2 each of length 0.50 m.

The combination of masses is made to swing in a vertical circle with both strings always in a straight line, as shown in Fig. 1.1.

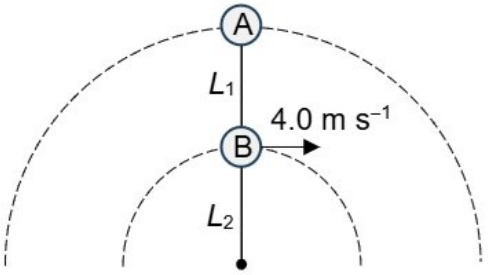


Fig 1.1

At the top of its motion, mass B is travelling at 4.0 m s^{-1}.

(a) Determine the speed of mass A at this instant.

speed of mass A = ... m s^{-1} [2]

(b) Show that the tension in L_1 at this instant is 200 N.

[1]

(c) Hence or otherwise, determine the tension in L_2 at this instant.

tension in L_2 = ... N [2]

(d) At another instant in time, the combination of masses is at the bottom of its motion as shown in Fig. 1.2.

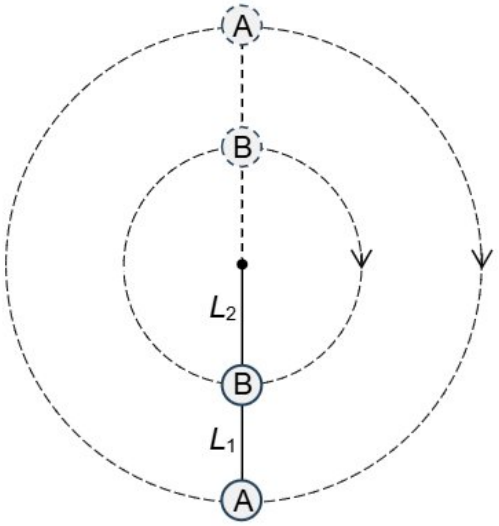


Fig 1.2

A student suggests that because the tensions in strings L_1 and L_2 change continuously during the motion, the total mechanical energy (i.e. kinetic and potential energy) of the system is not conserved.

With reference to tensions in the strings, state whether you agree with this statement and explain your reasoning.

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.....[2]

[Total: 7]